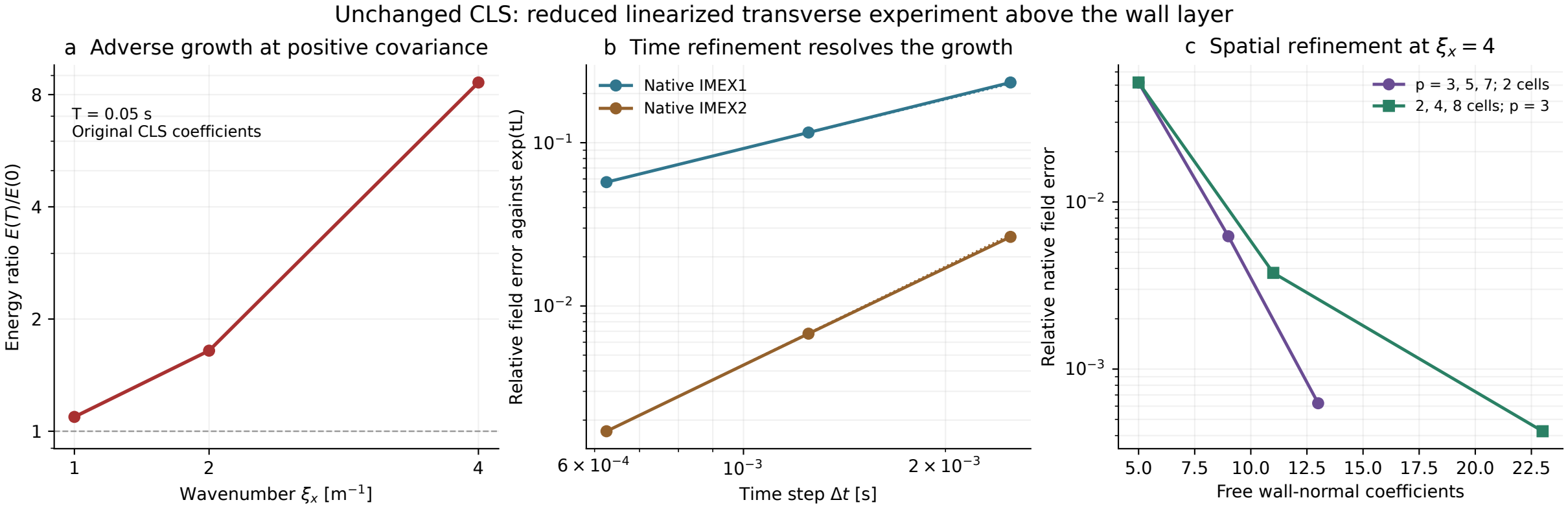
Unchanged CLS: reduced linearized transverse experiment above the wall layer
a Adverse growth at positive covariance
T = 0.05 s
Original CLS coefficients
Energy ratio E(T)/E(0)
Wavenumber ξx [m−1]
b Time refinement resolves the growth
Native IMEX1
Native IMEX2
Relative field error against exp(tL)
Time step Δt [s]
c Spatial refinement at ξx = 4
p = 3, 5, 7; 2 cells
2, 4, 8 cells; p = 3
Relative native field error
Free wall-normal coefficients